==========

==========

CHAPTER 15 — MEMORY TYPES AND CACHEABILITY

==========

==========

A maximum-detail, RTOS/QNX-style technical breakdown of ARMv8 memory types, cacheability rules, ordering semantics, device vs normal memory behavior, and how MAIR/TCR/descriptors interact to produce final memory attributes.

****

****

15.1 — Introduction to Memory Types

****

****

ARMv8 defines two fundamental memory categories:

- **Normal Memory**
- **Device Memory**

These categories determine:

- cacheability
- ordering rules
- speculation behavior
- prefetch behavior
- alignment requirements
- atomic operation legality
- DMA interaction
- TLB behavior

Memory type is determined by:

- **AttrIndx** in descriptors
- **MAIR_EL1 / MAIR_EL2** entries
- **TCR_ELx** overrides
- **TTBR_ELx** overrides

This chapter explains each memory type in maximum detail.

****

****

15.2 — Normal Memory

****

****

Normal memory is used for:

- RAM
- code
- data
- stacks

- heap
- DMA buffers
- shared memory

Normal memory supports:

- caching
- speculation
- prefetching
- out-of-order execution
- atomic operations
- SMP coherence

Normal memory is the default for most mappings.

****

****

15.3 — Device Memory

****

****

Device memory is used for:

- MMIO registers
- PCIe BARs
- interrupt controllers
- UARTs
- timers
- GPIO
- DMA control registers

Device memory is:

- **non-cacheable**
- **non-speculative**
- **strongly ordered**
- **non-prefetchable**
- **non-coherent**
- **alignment-sensitive**

Device memory must be mapped correctly to avoid hardware malfunction.

****

****

15.4 — MAIR_ELx: Memory Attribute Indirection Register

****

****

MAIR_EL1 and MAIR_EL2 define **8 memory types**.

Each entry is 8 bits:

```
MAIR_ELx = [Attr7][Attr6]...[Attr0]
```

AttrIndx in descriptors selects one of these entries.

****

****

15.5 — Normal Memory Encodings

****

****

Normal Write-Back (WB)

Most common for RAM.

```
0xFF = Inner/Outer Write-Back, Read/Write Allocate
```

Normal Write-Through (WT)

Used rarely.

```
0xBB = Inner/Outer Write-Through
```

Normal Non-Cacheable (NC)

Used for DMA buffers or uncached regions.

```
0x44 = Non-cacheable
```

Teaching Note

Normal memory supports speculation and prefetch.

****

****

15.6 — Device Memory Encodings

****

****

Device-nGnRnE (Strongest Ordering)

Most restrictive, safest for MMIO.

`0x00 = Device, non-Gathering, non-Reordering, non-Early write acknowledgment`

Device-nGnRE

Allows early write acknowledgment.

`0x04 = Device, non-Gathering, non-Reordering, Early write acknowledgment`

Device-nGRE

Allows limited reordering.

`0x08 = Device, non-Gathering, Reordering, Early write acknowledgment`

Teaching Note

Device-nGnRnE is recommended for all MMIO.

****

****

15.7 — Gathering, Reordering, Early Write Ack

****

****

Device memory attributes control:

Gathering

Combining multiple writes into one transaction.

Reordering

Allowing writes to be reordered.

Early Write Acknowledgment

CPU receives “write complete” before device processes it.

Rules

- MMIO registers must be **non-gathering**.
- MMIO registers must be **non-reordering**.
- MMIO registers should be **non-early-ack** unless hardware requires otherwise.

****

****

15.8 — Cacheability Rules

****

****

Normal Memory

- can be cached
- can be prefetched
- can be speculated
- can be reordered
- supports atomic operations

Device Memory

- must not be cached
- must not be prefetched
- must not be speculated
- must not be reordered
- atomic operations may be illegal

Implementation Note

Incorrect cacheability causes catastrophic hardware malfunction.

****

****

15.9 — Shareability Rules

****

****

Shareability determines SMP coherence.

Inner Shareable

- required for SMP
- ensures L1/L2 coherence

Outer Shareable

- used for multi-cluster systems

Non-Shareable

- used for private memory
- used for device memory

Teaching Note

Normal memory should be Inner Shareable.

****

****

15.10 — Ordering Rules

****

****

Normal Memory

- weakly ordered
- loads/stores may reorder
- speculation allowed
- prefetch allowed

Device Memory

- strongly ordered
- loads/stores cannot reorder
- speculation forbidden
- prefetch forbidden

Implementation Note

Device memory ordering is critical for correct hardware behavior.

****

****

15.11 — Speculation Rules

****

****

Normal Memory

- speculative loads allowed
- speculative stores allowed
- speculative execution allowed

Device Memory

- speculative loads forbidden
- speculative stores forbidden
- speculative execution forbidden

Teaching Note

Speculation on device memory causes undefined behavior.

****

****

15.12 — Prefetch Rules

****

****

Normal Memory

- prefetch allowed
- improves performance

Device Memory

- prefetch forbidden
- may corrupt device state

Implementation Note

Prefetchers must not touch MMIO.

****

****

15.13 — Atomic Operation Rules

****

****

Normal Memory

- atomic operations allowed
- required for locks

Device Memory

- atomic operations may be illegal
- may cause bus errors
- may corrupt device state

Teaching Note

Never use atomics on MMIO.

****

****

15.14 — DMA Interaction Rules

****

****

DMA Buffers

Should be Normal WB or Normal NC.

MMIO Registers

Must be Device-nGnRnE.

Cache Maintenance

Required before/after DMA:

- **DC CVAC**
- **DC CIVAC**
- **DSB SY**

Implementation Note

Incorrect DMA attributes cause data corruption.

****

****

15.15 — Descriptor Attribute Selection

****

****

Descriptor contains:

`AttrIndx[2:0]`

This selects MAIR entry.

Example

```
AttrIndx = 0 → Normal WB
AttrIndx = 1 → Normal NC
AttrIndx = 2 → Device-nGnRnE
```

Teaching Note

AttrIndx must match memory type.

****

****

15.16 — Stage-1 vs Stage-2 Attribute Merging

****

****

Final memory type = Stage-2 overrides Stage-1.

Example

- Stage-1: Normal WB
- Stage-2: Device-nGnRnE

Final = Device-nGnRnE

Hypervisor always wins.

****

****

15.17 — TCR/TTBR Attribute Overrides

****

****

TTBR and TCR can override descriptor attributes:

- IRGN → inner cacheability
- ORGN → outer cacheability
- SH → shareability

Rules

- Overrides apply only to page table walks.
- Overrides do not change final memory type.

****

****

15.18 — Common Attribute Mistakes

****

****

Mistake 1 — Mapping MMIO as Normal WB

Causes device malfunction.

Mistake 2 — Mapping RAM as Device

Disables caching → severe performance loss.

Mistake 3 — Using WT instead of WB

Reduces performance.

Mistake 4 — Using NC for code

Breaks instruction cache behavior.

Mistake 5 — Using Device for DMA buffers

Breaks DMA.

****

****

15.19 — Full Attribute Example

****

****

Given descriptor:

```
AttrIndx = 2
```

MAIR_EL1[2] = 0x00 → Device-nGnRnE

Final memory type:

- non-cacheable
- non-speculative
- strongly ordered
- no prefetch
- no atomics

Used for MMIO.

****

****

15.20 — Chapter Summary

****

****

This chapter covered:

- normal vs device memory
- MAIR encodings
- cacheability rules
- ordering rules
- speculation rules
- prefetch rules
- atomic operation rules
- DMA interaction
- descriptor attribute selection
- Stage-1 vs Stage-2 attribute merging
- TCR/TTBR overrides
- common mistakes
- full examples